

Lab 1 - Requirements Table

Hospital Bed & ICU Allocation Dashboard

Scenario basis: Emergency room bed management dashboard with real-time ward occupancy, triage-based prioritization, and inter-facility ambulance transfer workflows.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	High	Pass: A higher-acuity patient is placed ahead of a lower-acuity patient when both are waiting. Fail: FIFO order leaves the higher-acuity patient behind.	Ensures emergency cases receive priority based on triage severity.
FR-002	Functional	The system shall display current bed status for monitored wards using Occupied, Cleaning, and Available states.	High	Pass: When a bed status changes, the new state is shown on the connected ER dashboard. Fail: The dashboard shows a previous state after the update.	Provides current occupancy information for bed allocation decisions.
FR-003	Functional	The system shall allow a Triage Nurse to enter patient intake vitals and use them to compute the patient's triage severity score.	High	Pass: After valid intake vitals are submitted, a triage score is produced and attached to the patient record. Fail: No score is produced from a completed intake.	Supplies the clinical input required for severity-based prioritization.
FR-004	Functional	The system shall prioritize patients requiring ICU transfer based on their triage severity score and identify them for transfer handling.	High	Pass: When a patient meets the configured ICU-transfer priority condition, the patient is flagged for ICU transfer handling. Fail: A qualifying patient remains unflagged.	Connects triage prioritization with ICU transfer workflow.
FR-005	Functional	The system shall create and track an inter-facility ambulance transfer workflow from transfer request through transfer confirmation.	Medium	Pass: A transfer record shows request status and is updated when transfer confirmation is received. Fail: The transfer cannot be tracked to a confirmation state.	Supports the required inter-facility ambulance coordination process.
NFR-001	Nonfunctional	The system shall propagate bed status transitions (Occupied, Cleaning, Available) to all connected ER dashboard clients in under 1 second via WebSockets.	High	Pass: Benchmarking tests show propagation below 1 second under the simulated peak-load test. Fail: Propagation reaches or exceeds 1 second.	Keeps the dashboard sufficiently real-time for bed management.
NFR-002	Nonfunctional	The system shall allow only authorized users to update bed status, triage information, or transfer workflow records and shall log rejected access attempts.	High	Pass: An unauthorized update is rejected and a corresponding access event is recorded. Fail: An unauthorized user can modify protected records or no rejection event is logged.	Protects operational and patient-related data from unauthorized changes.

Actors reflected in the UML: Triage Nurse, Hospital Administrator, and Inter-facility Ambulance Service. The first two are the stakeholders/actors named in the problem statement; the ambulance service is represented as an external actor because the scenario explicitly includes inter-facility ambulance transfer orchestration.